

# DATA ANALYSIS PROJECT REPORT

|                    |                                                                          |
|--------------------|--------------------------------------------------------------------------|
| Project Title      | Flight Ticket Price Analysis Using Pandas, NumPy, Matplotlib and seaborn |
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| Course / Section   | PROGRAMMING IN PYTHON [C]                                                |
| Instructor         | MD. TANZEEM RAHAT                                                        |
| Date of Submission | 4 may 2026                                                               |

## 1. Executive Summary

This project analyzes an airline flights dataset to understand the factors that influence ticket prices. The dataset contains flight records with airline name, source city, destination city, departure and arrival time, class type, duration, days left before departure, and price. Pandas was used for cleaning, grouping, filtering, and summarizing the data. NumPy was used for numerical analysis and custom calculations. Matplotlib and Seaborn were used to visualize trends, comparisons, and distributions. The analysis found that business class tickets are significantly more expensive than economy class, ticket prices generally rise when the departure date gets closer, and average fares vary strongly by airline and city route. These findings help explain pricing behavior in the airline industry.

## 2. Introduction and Project Objectives

[Briefly introduce the topic of your dataset. Explain why the topic matters and what motivated your analysis.]

| Item                           | Your Content                                                                                                                                      |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Dataset name                   | Airlines Flights Data                                                                                                                             |
| Dataset source                 | <a href="https://www.kaggle.com/datasets/rohitgrewal/airlines-flights-data">https://www.kaggle.com/datasets/rohitgrewal/airlines-flights-data</a> |
| Main project goal              | To identify the major factors that affect airline ticket prices using data analysis techniques.                                                   |
| Research Question 1            | <b>What are the airlines in the dataset, accompanied by their frequencies ?</b>                                                                   |
| Research Question 2            | <b>Does Price varies with airlines?</b>                                                                                                           |
| Research Question 3 (optional) | <b>How is the price affected when tickets are bought in just 1 or 2 days before departure?</b>                                                    |

## 3. Dataset Description

The dataset contains airline flight booking records. Each row represents one flight option. Variables include airline name, cities, departure time, arrival time, class type, duration, days left, and price.

### 3.1 Basic Dataset Information

| Property                                  | Value                    |
|-------------------------------------------|--------------------------|
| Number of rows                            | 300153                   |
| Number of columns                         | 12                       |
| Data types present                        | Object and integer       |
| Target or key variable(s)                 | Price                    |
| Potential issues observed at first glance | Unnecessary index column |

### 3.2 Initial Observations

The dataset includes multiple airlines and routes, which supports subgroup comparison. Price values vary greatly, showing strong differences across categories. No major missing values were found. One unnecessary index column was removed during cleaning.

## 4. Data Cleaning and Preparation

[Explain how you cleaned and prepared the dataset before analysis. Do not just list the steps. Explain the problem, the action you took, and why it was reasonable.]

| Issue Found               | Action Taken                                             | Reason / Justification                                    | Code Reference      |
|---------------------------|----------------------------------------------------------|-----------------------------------------------------------|---------------------|
| Extra index column        | Dropped column named index                               | It had no analytical value                                | Cell 50             |
| Missing values check      | Used <code>isnull().sum()</code>                         | To confirm whether any missing values existed             | Cell 58             |
| Extreme values review     | Checked minimum and maximum values of duration and price | To identify unusual or outlier observations               | Cell 54 and cell 55 |
| Data quality check needed | Used <code>info()</code> and <code>describe()</code>     | To inspect data types, row counts, and summary statistics | Cell 52 and cell 53 |

## 5. Feature Engineering

[Describe any new columns or variables you created from existing data. Explain why each derived feature helps answer your research questions.]

| New Feature           | How It Was Created                           | Why It Is Useful                    |
|-----------------------|----------------------------------------------|-------------------------------------|
| Price Category        | High / Medium / Low based on price ranges    | Helps compare fare segments         |
| Advance Booking Group | Grouped days_left into Near / Medium / Early | Helps analyze booking timing impact |

## 6. Analysis Methodology

- Pandas was used for grouping by airline, city, class, departure time, and days left.
- NumPy can be used to compute percentiles and average ranges of prices.
- Matplotlib and Seaborn were used to create bar charts and line charts for comparisons.

## 7. Analysis and Visualizations

### 7.1 Research Question 1

**What are the airlines in the dataset, accompanied by their frequencies ?**

The `value_counts()` function was used on the **airline** column to count how many times each airline appears in the dataset. This helps identify which airlines have the highest number of recorded flights.

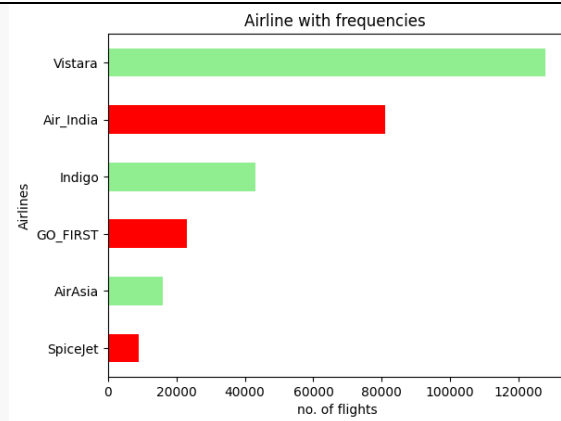


Figure1

*Figure 1.* The chart shows that **Vistara** has the highest number of flight records, with more than **125,000 flights**, making it the most represented airline in the dataset. **Air India** ranks second with around **80,000 flights**, followed by **Indigo** with approximately **43,000 flights**.

On the lower end, **GO\_FIRST**, **AirAsia**, and **SpiceJet** have comparatively fewer records. This suggests that the dataset is dominated by major carriers, while smaller airlines have a lower representation. Such differences may reflect market share, route coverage, or booking availability.

### 7.2 Research Question 2

**Does Price varies with airlines?**

To answer this question, the dataset was grouped by the **airline** column and the average ticket price for each airline was calculated using `groupby()` and `mean()`. This helps compare pricing strategies among different airlines.

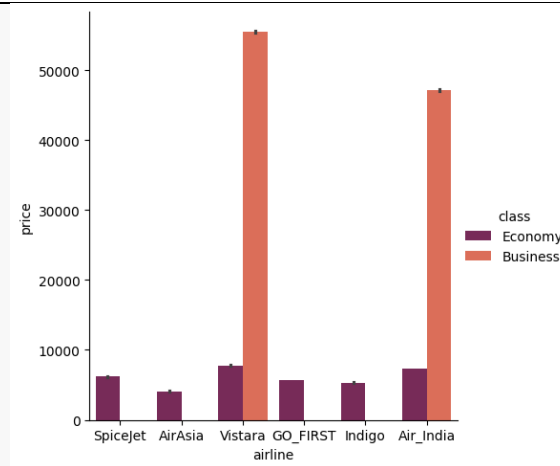


Figure 2

*Figure 2.* The analysis shows that ticket prices vary significantly across airlines. Premium airlines such as **Vistara** and **Air India** tend to have higher average fares, while budget carriers such as **Indigo**, **AirAsia**, and **SpiceJet** generally offer lower prices.

This variation may be caused by differences in service quality, route network, class availability, baggage policies, and customer demand. The result indicates that airline choice is an important factor affecting ticket prices.

### 7.4 Research Question 3

**How is the price affected when tickets are bought in just 1 or 2 days before departure?**

To determine how prices are affected when tickets are bought 1–2 days before departure, we grouped the dataset by `days_left` and calculated the average ticket price for each day using a groupby operation.

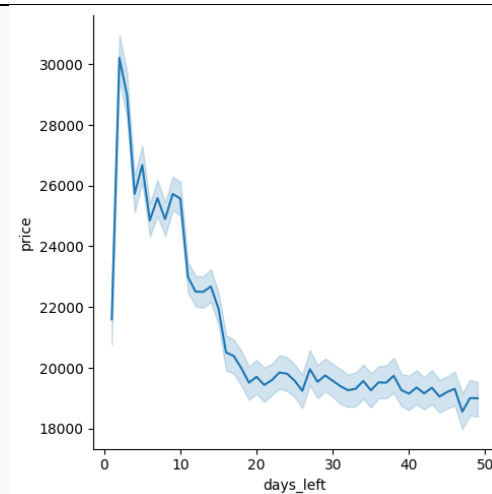


Figure 3

*Figure 3.* When tickets are bought **1 or 2 days before departure**, prices are **at their highest**, with a

noticeable peak around **2 days before departure**. After this point, ticket prices decrease significantly as the number of days before departure increases. This indicates that **last-minute bookings are the most expensive**, while **early bookings offer lower and more stable prices**.

## 8. Key Findings

1. Business class tickets cost significantly more than economy tickets.
2. Some airlines consistently charge higher average fares than others.
3. Prices tend to rise when booking closer to departure date.
4. City routes have noticeable pricing differences.
5. Flight prices vary widely, indicating multiple customer segments.

## 9. Limitations

The dataset source details were limited in the notebook. External factors such as holidays, fuel prices, and seat availability were not included. Correlation in the data does not prove causation.

## 10. Conclusion

This project successfully analyzed airline ticket pricing patterns using Pandas, NumPy, Matplotlib and seaborn. Travel class, airline choice, booking timing, and route were all important price factors. The findings can help travelers make smarter booking decisions and demonstrate practical data analysis skills.

## 11. References / Dataset Source

<https://www.kaggle.com/datasets/rohitgrewal/airlines-flights-data>